

Technical Document #307: Data Engineering - Comprehensive Overview

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Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Data warehouses store structured data optimized for analytical queries.
- Stream processing handles continuous data streams in real-time.
- Data lakes store raw data in its native format.